

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

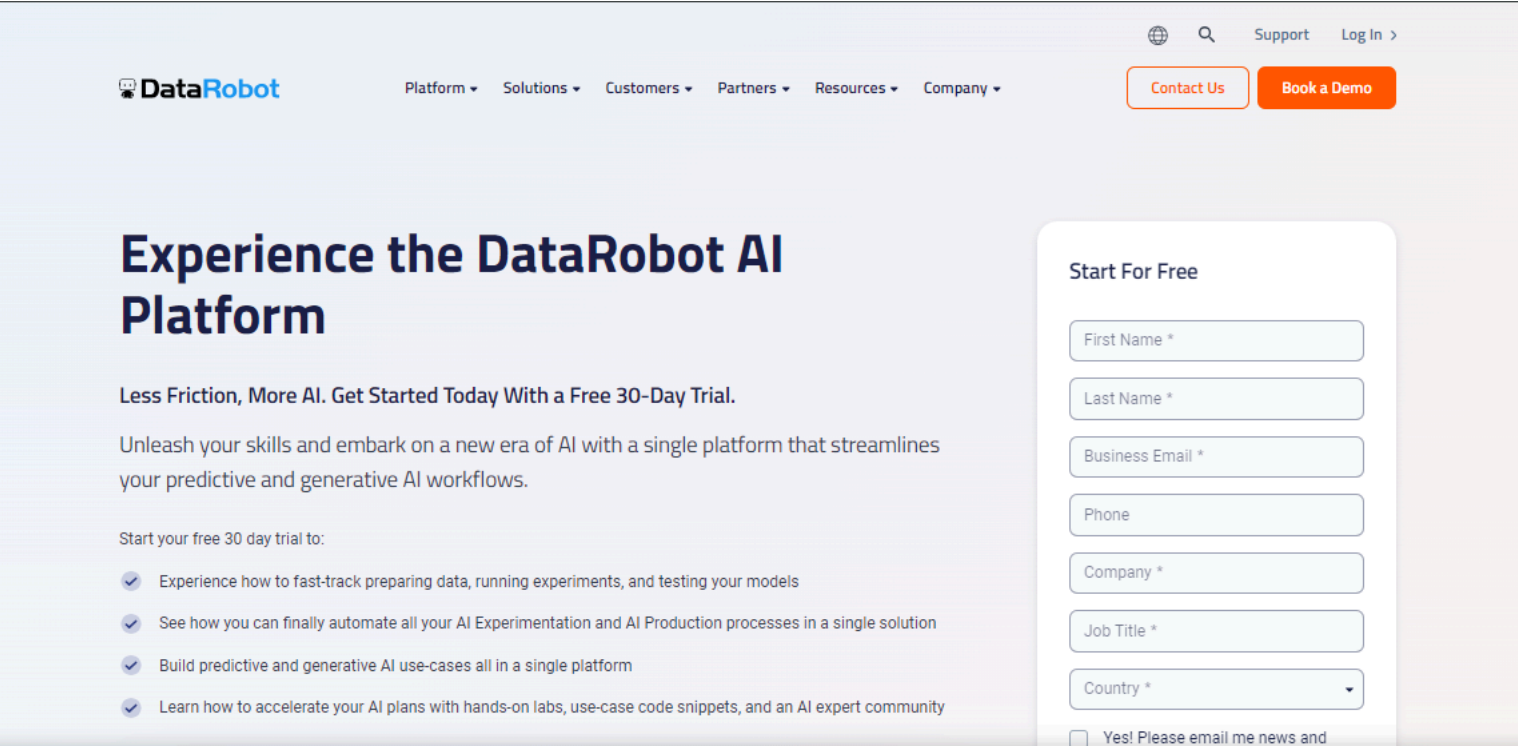
After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start for free" section and create an account.



Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

Step 4: A new window will open; select the relevant option for signing up.

Welcome back to DataRobot



Sign in with Google



Sign in with GitHub

or

Email

example@email.com

Password

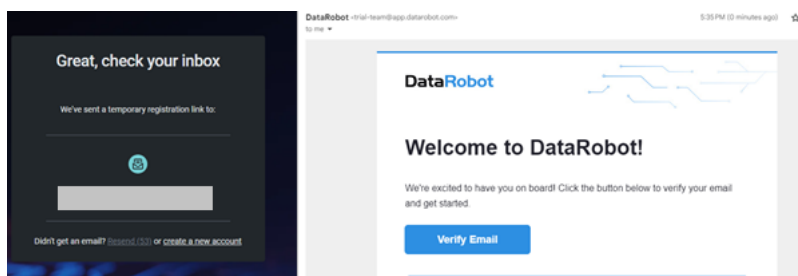
.....



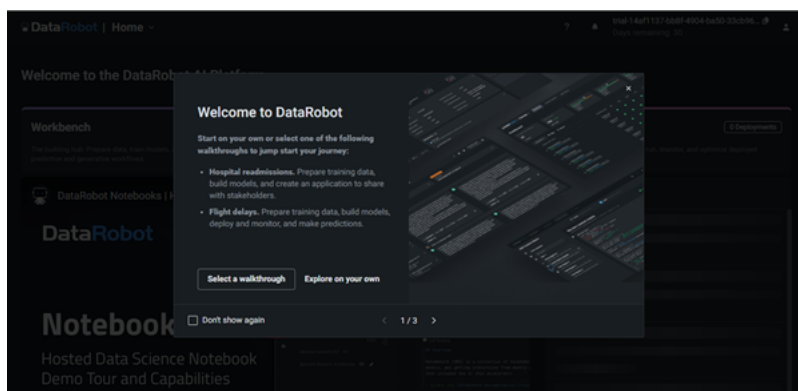
[Reset password](#)

Sign in

Step 5: Confirm your email by clicking **Verify Email** in your inbox.

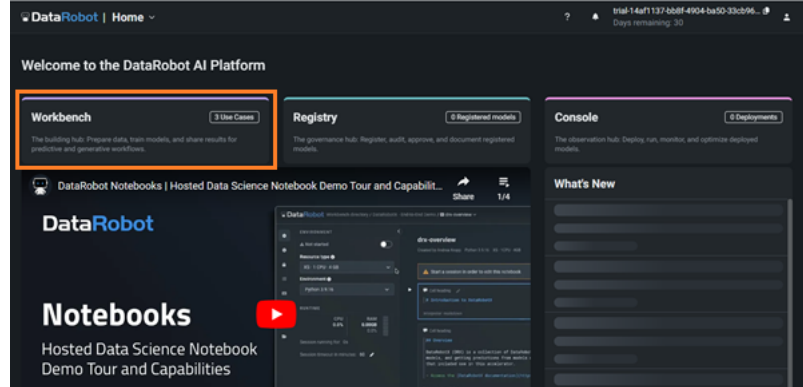


Step 6: Sign up and start your first experience of using the Generative AI tool.
The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

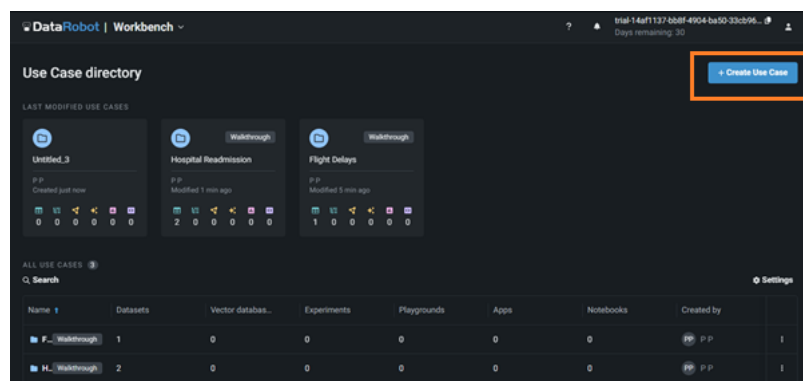


Task 2: Add a data set

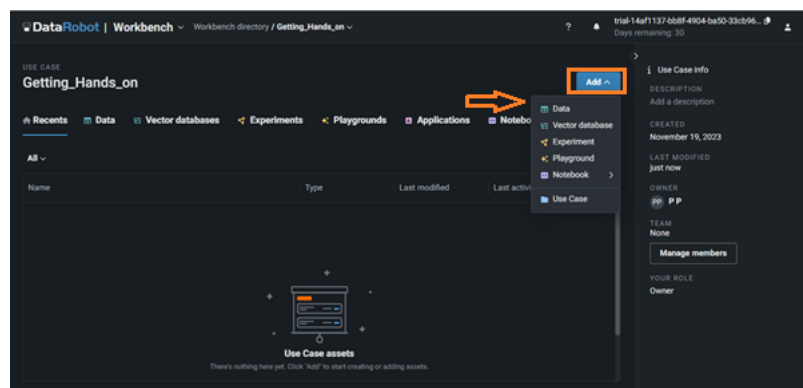
Step 7: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.



Step 8: Click **Create Use Case**.



Step 9: Click **Add** and **Data** to include the data set in your use case.



Step 10: **Upload** your data set or **Connect** to the data source; however, for this lab, you can select an in-built sample data set *HOSPITAL_READMISSION_DATA*.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e83-4e52-b732-acce

Days remaining: 29

Add data

Close

Preview

Add

DATA STORES

↑ Upload

+ Connect

Q Search

Data Registry

DataRobot

Demo Connection

Snowflake

Data Registry

Q Search

☐	Name	Created on ↑	Type	Last modified	Created by
☐	HOSPITAL_READMISSION_DATA	1d ago	Snapshot	1d ago	PV Pratiksh

< Previous

1

Next >

1 per page

1-1 of 1 items

Step 11: Once you select the data set, you can see a preview of it. You can also view the data set's features, as shown below. Click **Add to Use Case**.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e83-4e52-b732-acce

Days remaining: 29

Add data

Close

Add

DATA STORES

↑ Upload

+ Connect

Q Search

Data Registry

DataRobot

Demo Connection

Snowflake

HOSPITAL_READMISSION_DATA

Data Registry / HOSPITAL_READMISSION_...

Features

Data preview

race	gender	age	weight	admission_type_id	disch
Categorical	Categorical	Categorical	Categorical	Categorical	Catego
Caucasian	Male	[50-60)		Emergency	Disch
Caucasian	Female	[70-80)		Urgent	Disch
Caucasian	Male	[60-70)		Elective	
Caucasian	Male	[60-70)			Disch
Caucasian	Female	[40-50)		Not Available	Disch
Caucasian	Male	[90-100)		Emergency	Disch
Caucasian	Male	[70-80)		Urgent	Disch
Caucasian	Female	[80-90)		Urgent	Disch
Hispanic	Female	[50-60)		Emergency	Disch

Snapshot sample

51 features | 1,511 rows

Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.

DataRobot | Workbench

Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA

?

trial-c3398e81-5e83-4e52-b732-aca1c

Days remaining: 29

HOSPITAL_READMISSION_DATA

Jul 29th, 2024 10:39 AM Snapshot Data actions

Data preview

Features

Feature lists

Show insights

Show features from: All Features

+ Create feature list

race

Categorical

Caucasian74%

AfricanAmerican21%

Other6%

gender

Categorical

Female54%

Male46%

age

Categorical

[70-80]26%

[60-70]22%

Other52%

weight

Categorical

==Missing==96%

[75-100]2%

Other2%

admission_type

Categorical

Emergency49%

Urgent19%

Other32%

Caucasian

Male

[50-60]

Emergency

Discharged

Caucasian

Female

[70-80]

Urgent

Discharged

Caucasian

Male

[60-70]

Emergency

Discharged

Snapshot sample

51 features | 1,511 rows

Information

DATE ADDED

July 29, 2024

ADDED BY

Pratiksha Verma

FEATURES

51

ROWS

10,000

SIZE

5.48 MB

DATABASE

Snowflake / Demo Connection

SCHEMA

TRIAL_READONLY

TABLE

HOSPITAL_READMISSION_DATA

Dataset Versions

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench

Workbench directory / Untitled_4

?

trial-c3398e81-5e83-4e52-b732-aca1c

Days remaining: 29

Set up new experiment

Dataset

Target

Additional settings

Exit

< Back

Target feature

Select the feature to make predictions on.

payer_code

pioglitazone

race

readmitted

repaglinide

troglitazone

tolbutamide

tolazamide

time_in_hospital

Uniq...

Missi...

Mean

Std Dev

7

9592

-

-

35

Categorical

1

0

-

-

30

Categorical

2

0

-

-

36

Categorical

2

0

-

-

7

Numeric

14

0

4.43

3.021

Experiment summary

HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset

Name

HOSPITAL_READMISSION_DATA

Rows

10,000

Features

51

Target

No target selected

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench

Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca10
Days remaining: 29

Set up new experiment

Dataset

Target

Additional settings

Exit < Back

Target feature

Select the feature to make predictions on.

readmitted

Target type: Binary classification

Positive class: 0 1

Modeling mode

Set the mode used for selecting which blueprints to build when training models.

Quick Autopilot

Optimization metric

Set the metric used when training models to evaluate and optimize accuracy.

LogLoss (Accuracy) Recommended

Number of rows

6000

5000

4000

3000

2000

1000

0

False

True

Values of readmitted

Experiment summary

HOSPITAL_READMISSION_DATA - 2007-30 11:23:57

Dataset

NameHOSPITAL_READMISSION_DATA

Rows10,000

Features51

Target

Featurereadmitted

Target typeBinary classification

Positive class1

Modeling modeQuick Autopilot

Optimization metricLogLoss

Training feature listInformative Features

Partitioning

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

DataRobot | Workbench

Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca10
Days remaining: 29

Set up new experiment

Dataset

Target

Additional settings

Exit < Back Start

Data partitioning

Time series modeling Preview

Additional settings

Partitioning method

Select the method for assigning rows to partitions when training models.

Stratified sampling
Rows are assigned to ensure similar target distribution across each partition.

Validation type

Cross-validation

Trains models on a specified number of folds, maximizing data use but also increasing run time.

Training-validation-holdout

Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds

Enter a value from 2 - 50.

Holdout percentage

Set the subset of data that is unavailable during training and validation. Enter a value

Experiment summary

HOSPITAL_READMISSION_DATA - 2007-30 11:23:57

Dataset

NameHOSPITAL_READMISSION_DATA

Rows10,000

Features51

Target

Featurereadmitted

Target typeBinary classification

Positive class1

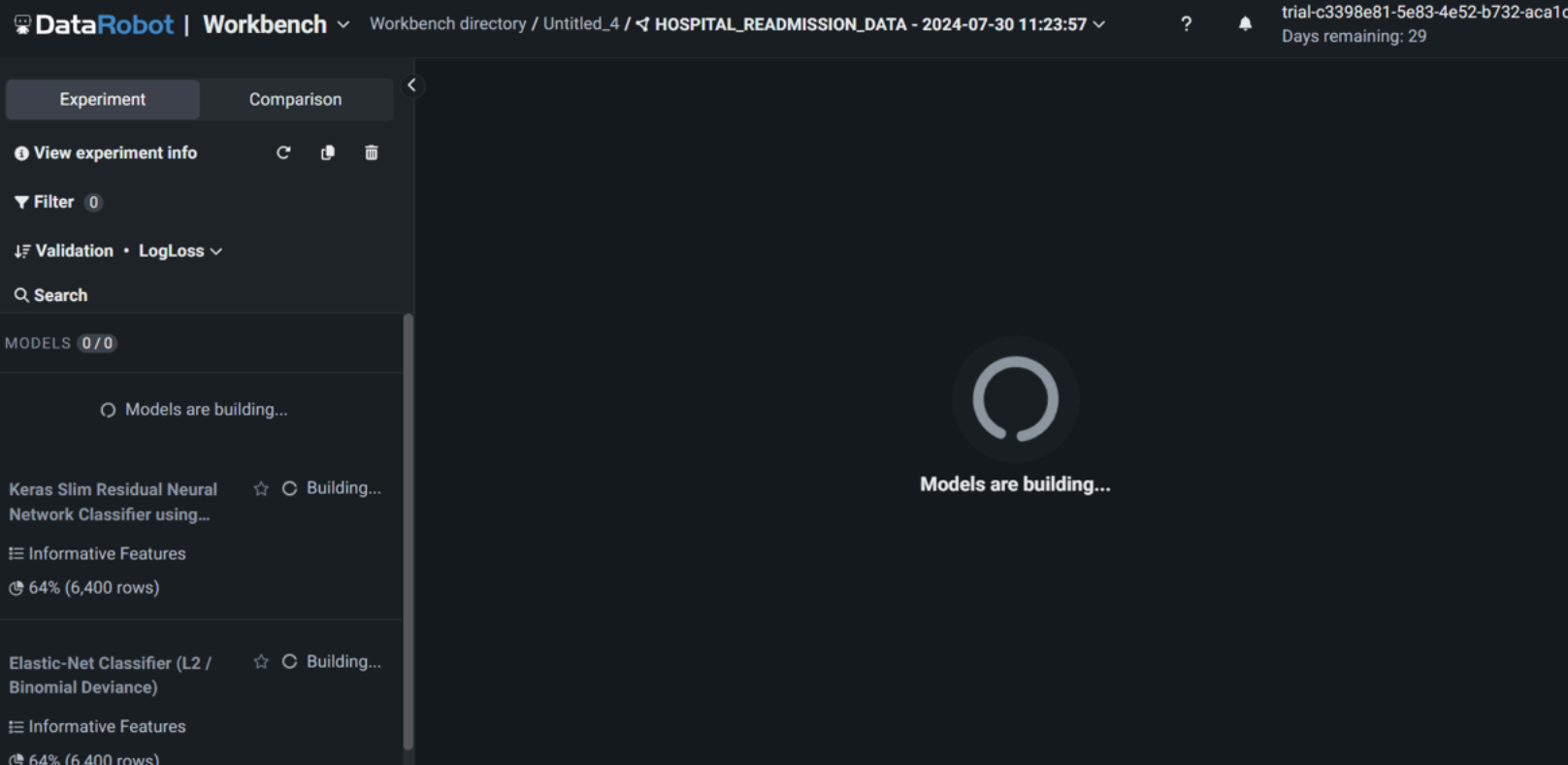
Modeling modeQuick Autopilot

Optimization metricLogLoss

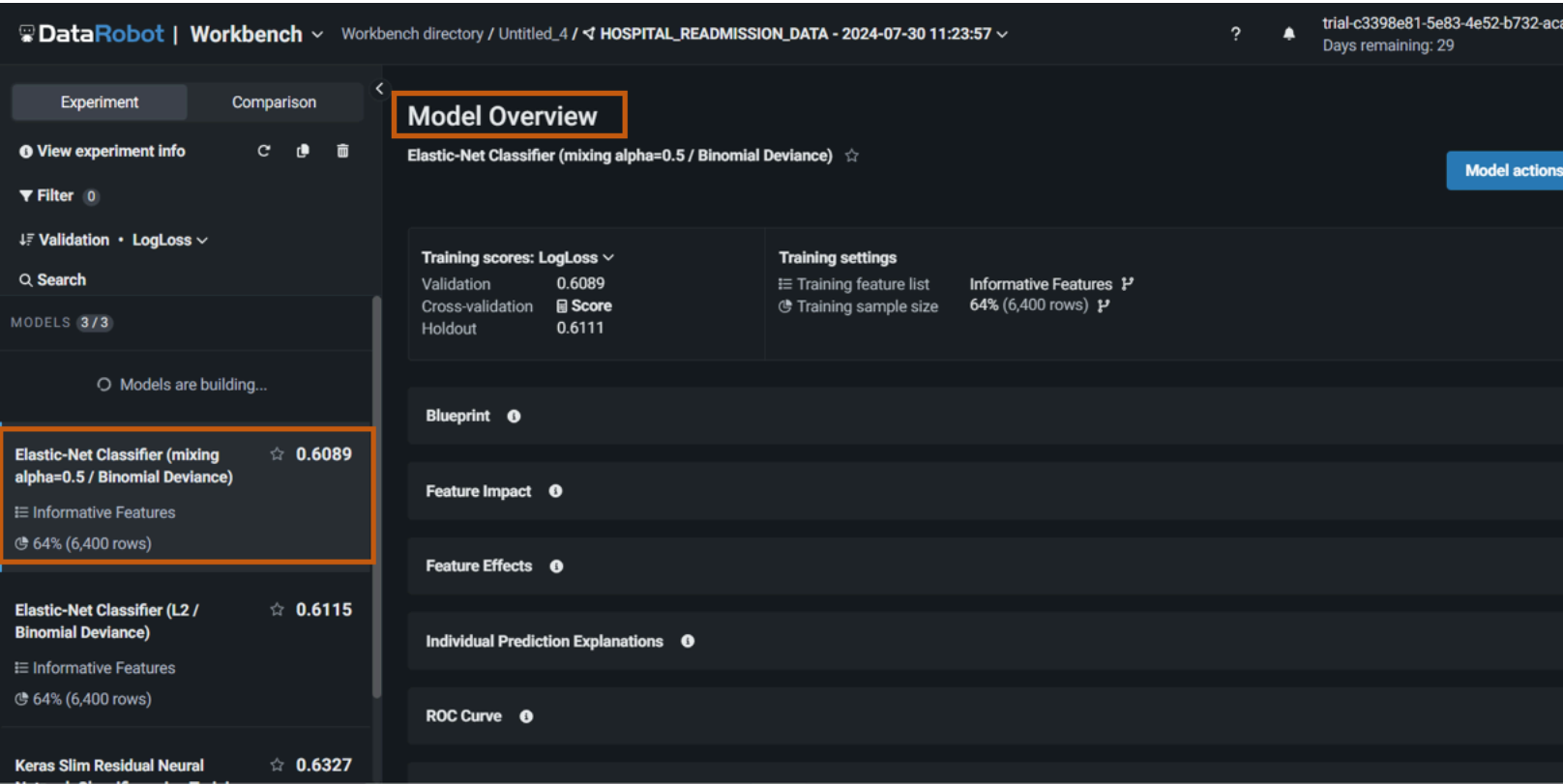
Training feature listInformative Features

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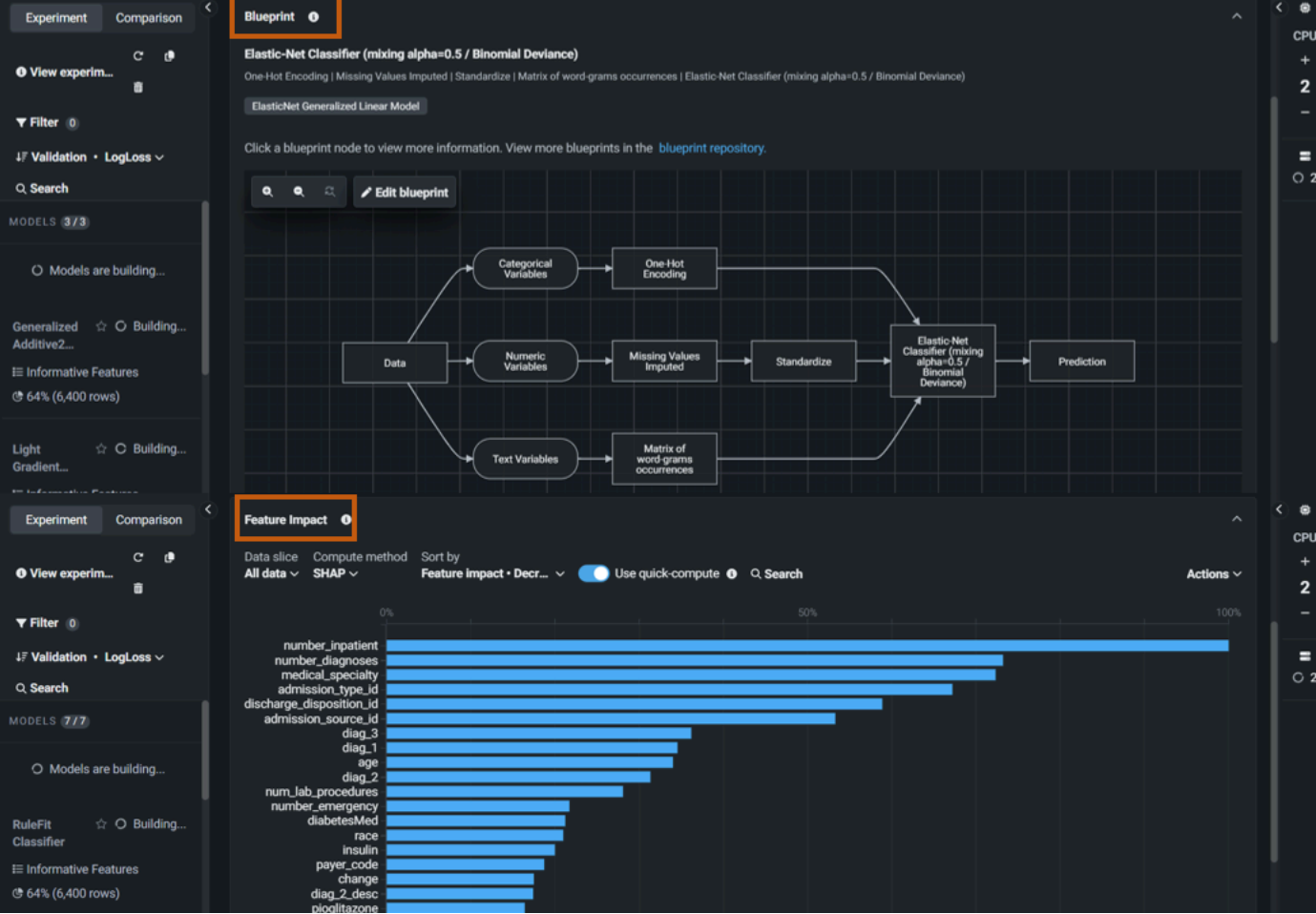
Step 18: Building models will take a while.



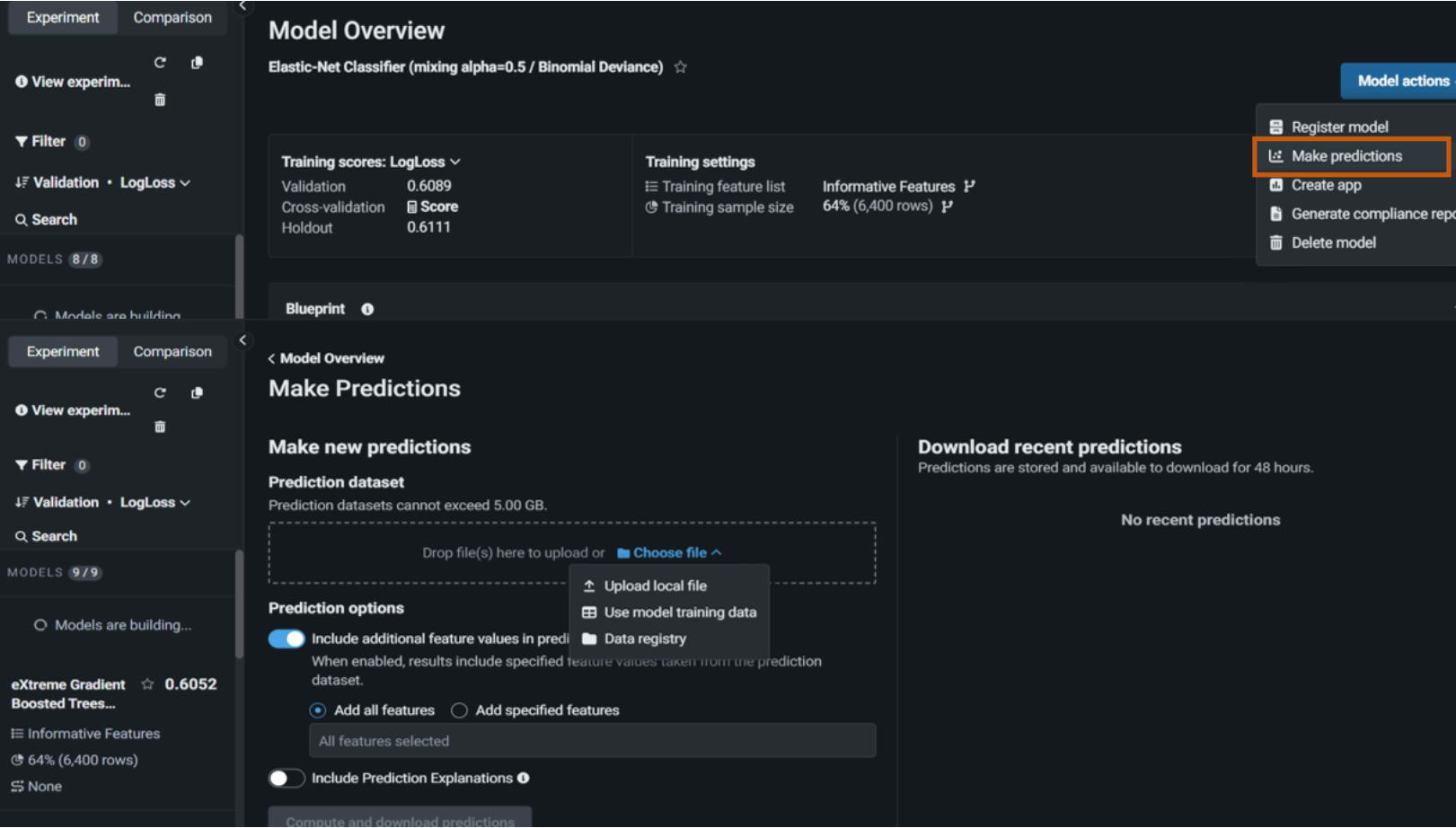
Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.



Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.



Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.



Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

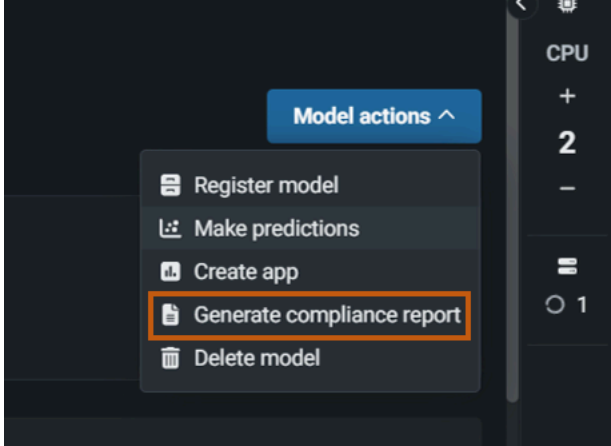


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Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

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